# Training Mechanics & Evaluation.

QLS–MiCM Workshop

# Why Evaluation Matters

Medical ML adds challenges:

- small datasets.
- class imbalance.
- noisy predictors.
- high stakes (false negatives matter).

**Accuracy alone is misleading.**

We need richer, more reliable evaluation methods.

# Cross-Validation for Model Selection

GridSearchCV:

- splits training data into folds.
- trains several models with different hyperparameters.
- selects the model with the best mean CV performance.
- avoids tuning on the test set

Core principle:

**Hyperparameters must be chosen using only training data.**

# What We Tune in This Exercise

- Architecture.
  - `hidden_layer_sizes`: (32,), (64,), (32,16)
- Regularization.
  - `alpha` → L2 penalty controlling model complexity
- Training Settings.
  - `max_iter`.
  - `random_state` (reproducibility).
  - optional: `early_stopping=True` to reduce overfitting + speed up training

The goal is to observe **how each change affects test performance**.

# Evaluation Metrics

From `classification_report`:

- **Accuracy**.
  - overall correctness; can be misleading with imbalance.
- **Precision**.
  - among predicted positives, how many are correct?
- **Recall (Sensitivity)**.
  - of true positives, how many were detected?
- **F1-score**.
  - harmonic mean of precision + recall; balances both.

Medical importance:

**Low recall = missed disease cases.**

# ROC AUC

If the model supports probabilities:

- ROC AUC measures ranking quality.
- threshold-independent.
- widely used in medical risk assessment.

High AUC means:

**Model gives higher scores to true positives more often.**

# Practical Training Notes

- Use `stratify=y` in train/test split to preserve class ratios.
- Use `n_jobs=-1` to speed up grid search.
- Watch for `ConvergenceWarning` -> increase `max_iter`.
- Don’t build huge grids - slow and prone to overfitting CV folds.